

test3- magnetism , A↓

⚠ This is a preview of the published version of the quiz

Started: Apr 7 at 12:27pm

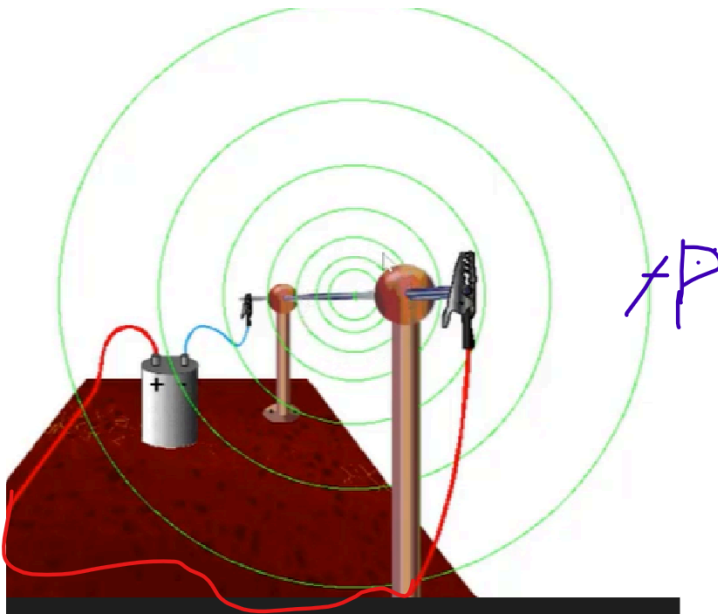
Quiz Instructions

You can not see the right answers. I WILL NOT REOPEN it if you mess up.

you can not see your wrong answers/ correct answers. think wisely. It is open note. open mind.



Question 1 5 pts



A current flows into the conductor (rod) that is perpendicular to your screen and connected to the battery shown (you see the positive terminal). A magnetic field is produced and you can see the lines in green.

A compass (small bar magnet) is placed at the location P. At P, The North side (red part) of the compass points

☐ up

☐ down

☐ left

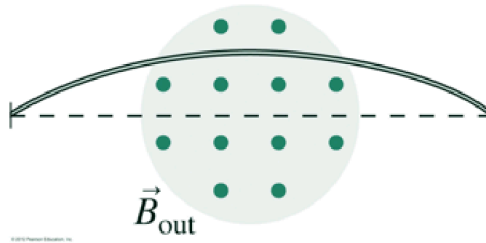


right



Question 2 5 pts

The figure shows a flexible conducting wire passing through a magnetic field that points out of the page. The wire is deflected upward, as shown. In which direction is current flowing in the wire?



to the left



can not be determined



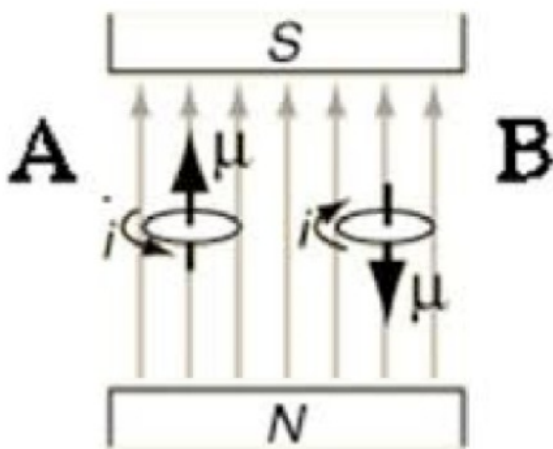
a magnetic field does not deflect



to the right



Question 3 5 pts



Here are magnetic dipoles inside a magnetic field. In Which positio, the dipole has the least PE (magnetic potential energy)?



B

☐

A

☐

neither

☐

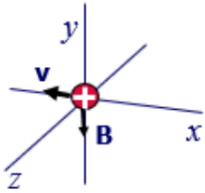
There is no such thing as a magnetic dipole

☐

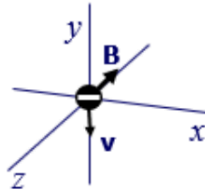
Question 4 5 pts

The figures shows five situations in which a charged particle with velocity $\mathbf{v}$ travels through a uniform magnetic field $\mathbf{B}$. In which situation, is the direction of the magnetic force along $+x$ axis ?

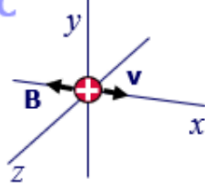
A



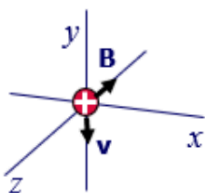
B



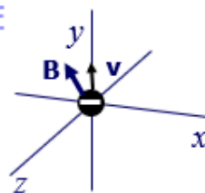
C



D



E

☐

B

☐

A

☐

D

☐

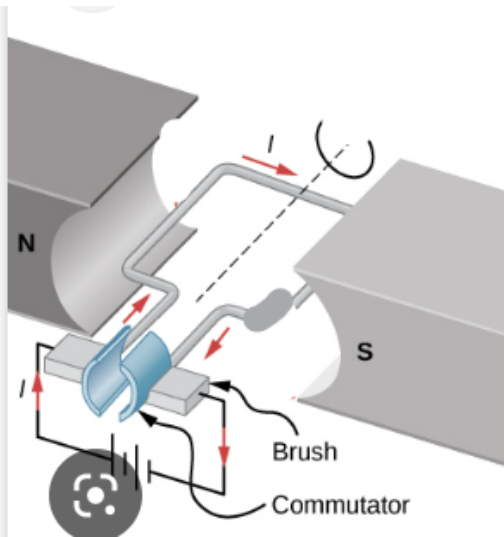
C

☐

E

☐

Question 5 5 pts

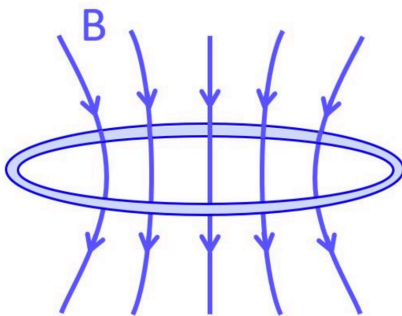


Here is a motor. You have a loop of current (so a magnetic dipole) inside a permanent magnetic field. The loop

- ☐ stays still
- ☐ is acted by a torque and start spinning CW
- ☐ starts oscillating about its position shown
- ☐ is acted by a torque and start spinning CCW



Question 6 5 pts



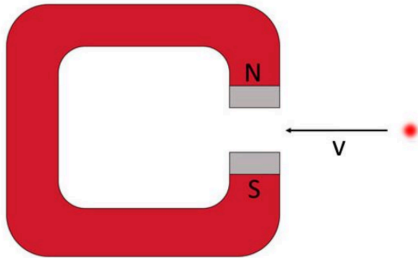
The current in the loop (its a conductor)

- ☐ flows CCW
- ☐ Idk
- ☐ can not be determined
- ☐ flows CW



Question 7 5 pts

2. What would be the direction of the force exerted on a proton moving as shown, by this magnetic field?



@ left



toward you (out of the screen)



@ right

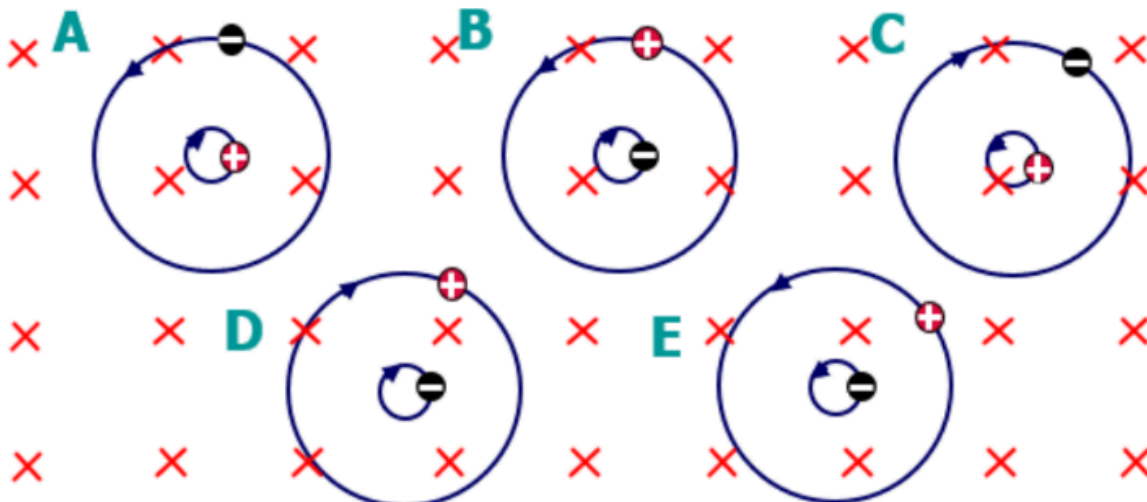


away from you (in the screen)



Question 8 5 pts

The figures shows the circular paths of two particles that travel at the same speed in a uniform magnetic field B , which is directed into the page. One particle is proton; the other is an electron (which is less massive). Which figure is physically reasonable?



hint: A positive charge moving up in a magnetic field going into the screen. which way is the force (we did it in class)

what about a negative charge. Also and the radius depends on the mass. an electron has less mass than a proton (2000 x less)

☐

C

☐

A

☐

E

☐

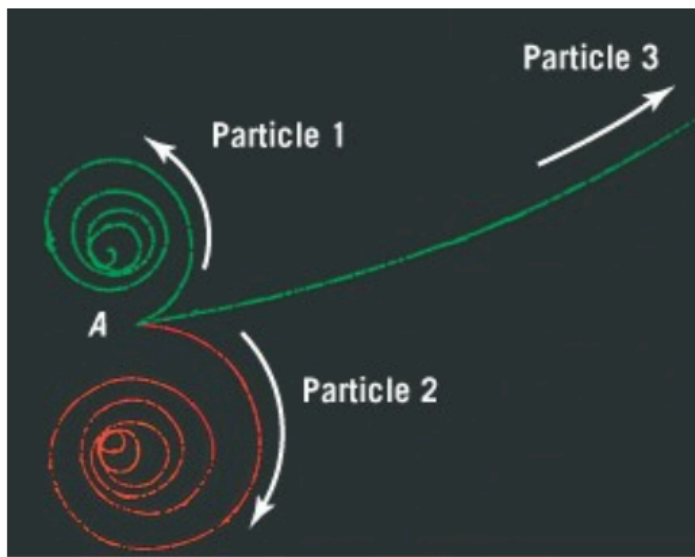
B

☐

D

☒

Question 9 5 pts



suppose the magnetic field is toward you (out of the screen) /

☐

particle 2 is negative and larger than 1

☐

particle 2 is positive and has the same mass as 1

☐

particle 3 has a positive charge

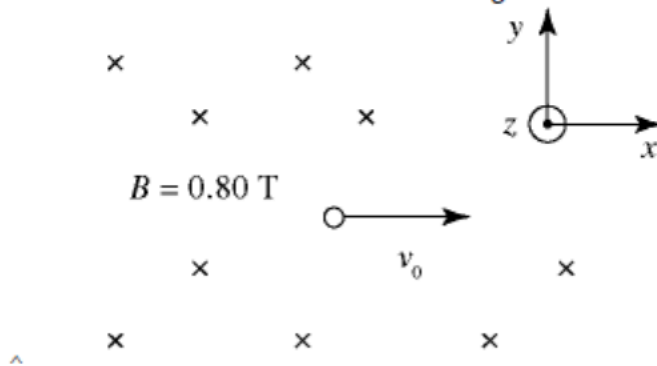
☐

particle 2 is positive and larger than 1

☒

Question 10 6.25 pts

- 1) A uniform magnetic field of magnitude 0.80 T in the negative z -direction is present in a region of space, as shown in the figure. A uniform electric field is also present. An electron that is projected with an initial velocity $v_0 = 9.1 \times 10^4\text{ m/s}$ in the positive x -direction passes through the region without deflection. What is the electric field vector in the region?



☐ 110kV down

☐ 73kV up

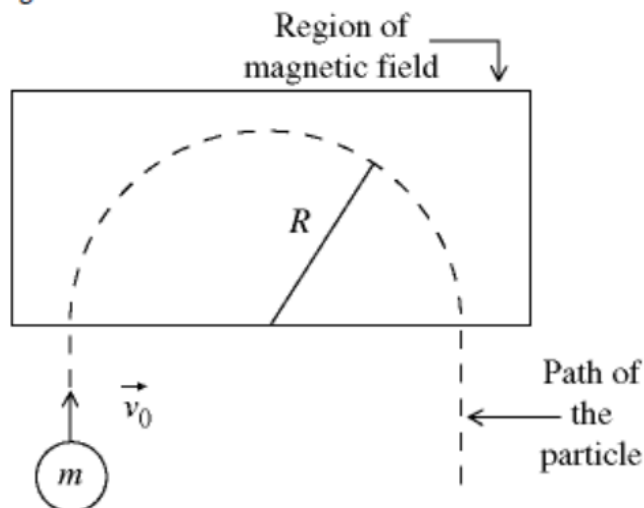
☐ 73kV down

☐ 110 kV up



Question 11 5 pts

- 12) As shown in the figure, a small particle of charge $q = -7.0 \times 10^{-6}\text{ C}$ and mass $m = 3.1 \times 10^{-12}\text{ kg}$ has velocity $v_0 = 9.4 \times 10^3\text{ m/s}$ as it enters a region of uniform magnetic field. The particle is observed to travel in the semicircular path shown, with radius $R = 5.0\text{ cm}$. Calculate the magnitude and direction of the magnetic field in the region.



☐ 0.083T out of the screen (toward you)

☐ 0.083T into the screen (away from you)



0.053 @ right



0.033 T out of the screen (toward you).



0.053 @ left



Question 12 5 pts

16) In a mass spectrometer, a singly-charged particle (charge e) has a speed of 1.0×10^6 m/s and enters a uniform magnetic field of 0.20 T. The radius of the circular orbit of the particle is 0.020 m. What is the mass of this particle? ($e = 1.60 \times 10^{-19}$ C)

3.2 10^{-28} kg3.1 10^{-31} kg6.4 10^{-28} kg1.7 10^{-27} kg4.5 10^{-27} kg

Question 13 5 pts

34) A circular coil of wire of 200 turns and diameter 2.0 cm carries a current of 4.0 A. It is placed in a magnetic field of 0.70 T with the plane of the coil making an angle of 30° with the magnetic field. What is the magnetic torque on the coil?

You need to draw and use geometry. See slide 19 of "big ideas"



0.40 Nm



0.088 Nm



0.29 Nm



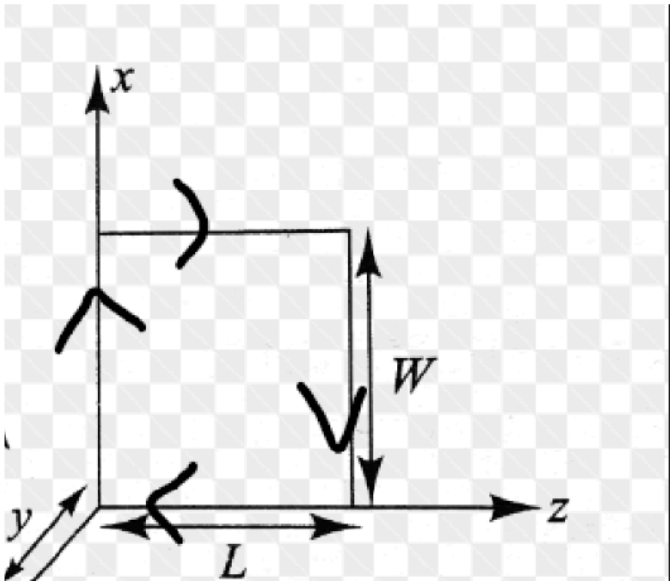
0.15 Nm



0.076 Nm



Question 14 5 pts



This is a loop of current (coil connected to a battery). It is equivalent to a bar magnet with a N and a S pole.

the magnetic moment dipole $\mathbf{u}$ is

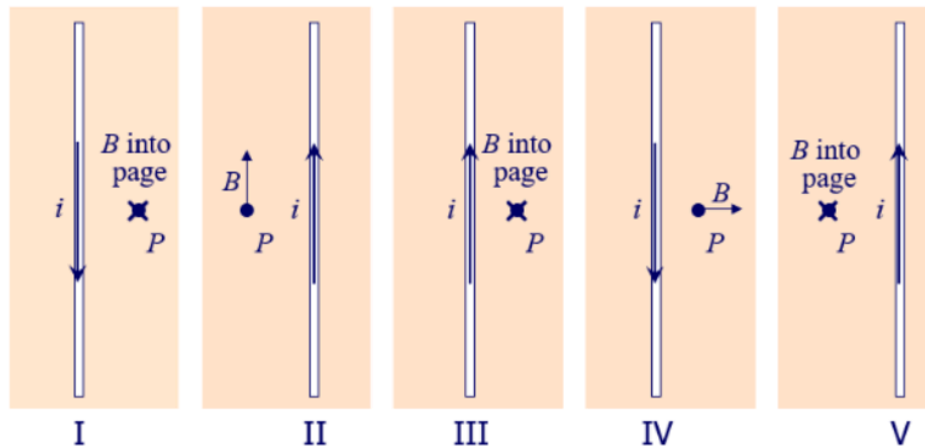
- ☐ along the negative x-axis (same direction)
- ☐ along the positive y-axis (same direction)
- ☐ along the negative y-axis (same direction)
- ☐ along the positive z-axis (same direction)
- ☐ along the positive x-axis (same direction)
- ☐ along the negative z-axis (same direction)



Question 15 5 pts

1. Which drawing below shows the correct direction of the magnetic field, B , at the point P ?

- A. I.
B. II.
C. III.
D. IV.



☐ IV

☐ I

☐ V

☐ II

☐ III

⋮

Question 16 3 pts

What is the main function of a cyclotron?

☐

To create magnetic fields

☐

To generate electricity

☐

To accelerate charged particles

☐

To amplify sound waves

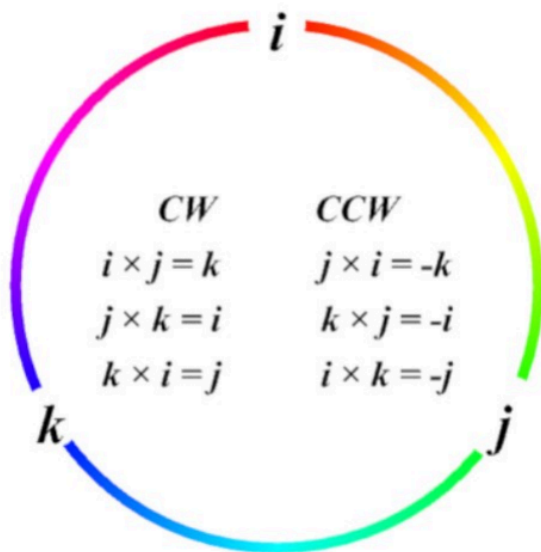
⋮

Question 17 6 pts

A 2.0-m wire carries a current of 15 A directed along the positive x axis in a region where the magnetic field is uniform and given by $B = (30\mathbf{i} - 40\mathbf{j})$ mT. What is the resulting magnetic force on the wire?

- a. $(+1.2 \mathbf{k})$ N
- b. $(-1.2 \mathbf{k})$ N
- c. $(-1.5 \mathbf{k})$ N
- d. $(+1.5 \mathbf{k})$ N
- e. $(+0.90 \mathbf{k})$ N

hint: cross product/ Its the magnetic force acting on a wire.


☐

b

☐

c

☐

e

☐

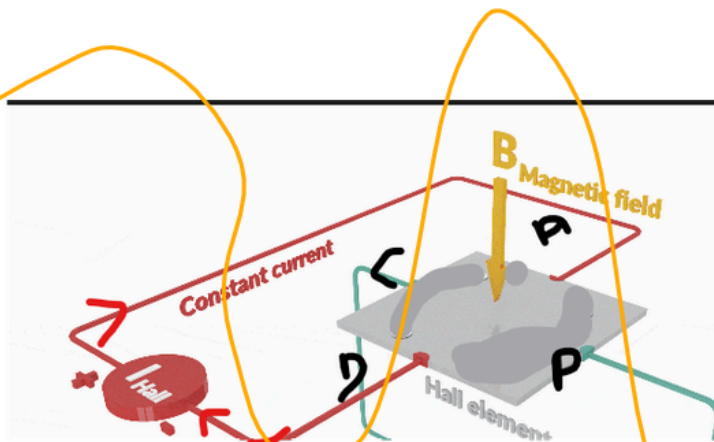
d

☐

a

☒

Question 18 5 pts



A current (red) flows from the battery (red +) through a conductive plate (grey) back to the battery. A magnetic field is introduced as shown (the direction is down). The positive charge (current) will be deflected toward

Don't pay attention to the yellow line.

☐

P

☐

A

☐

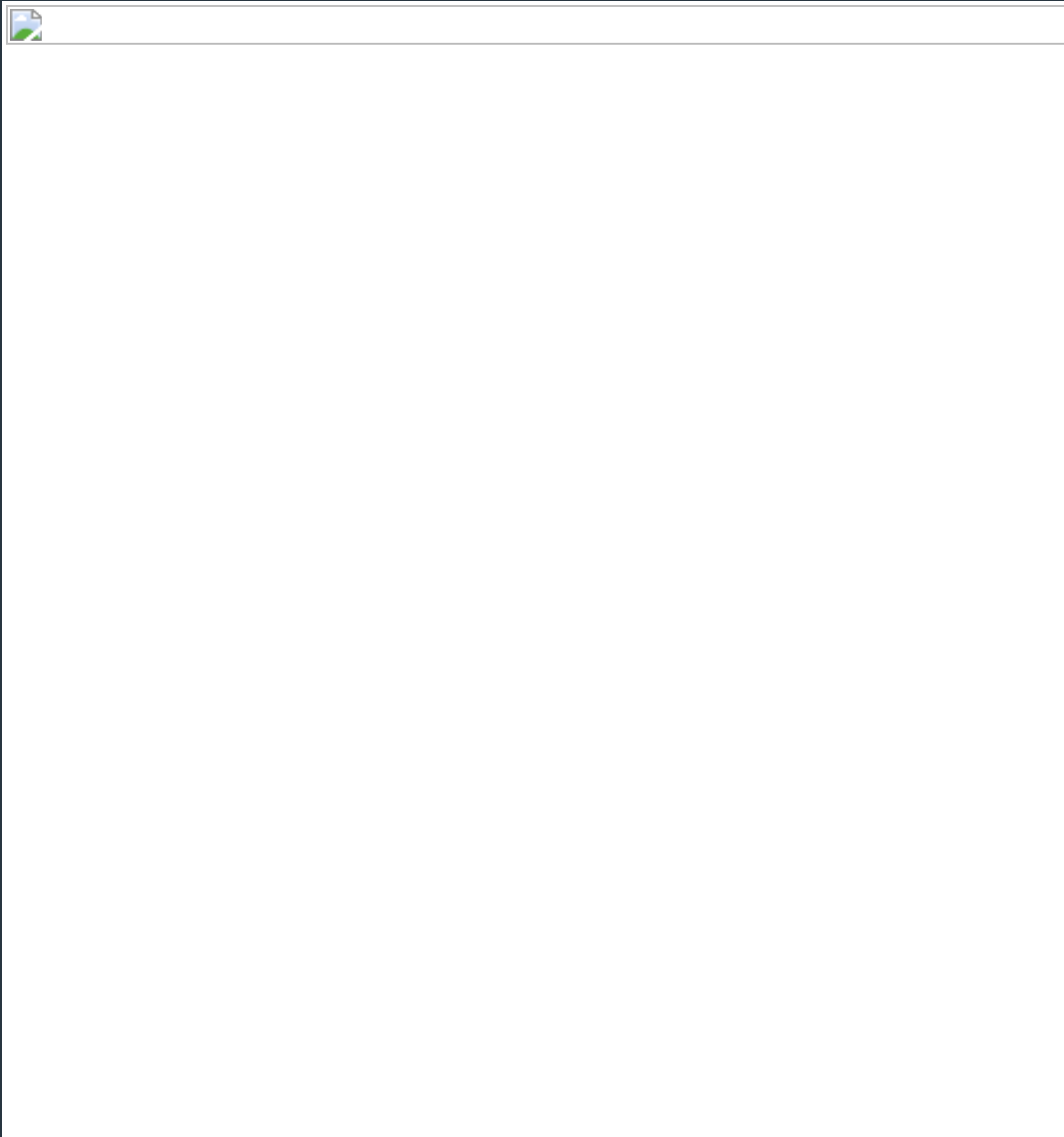
C

☐

will not be deflected

⋮

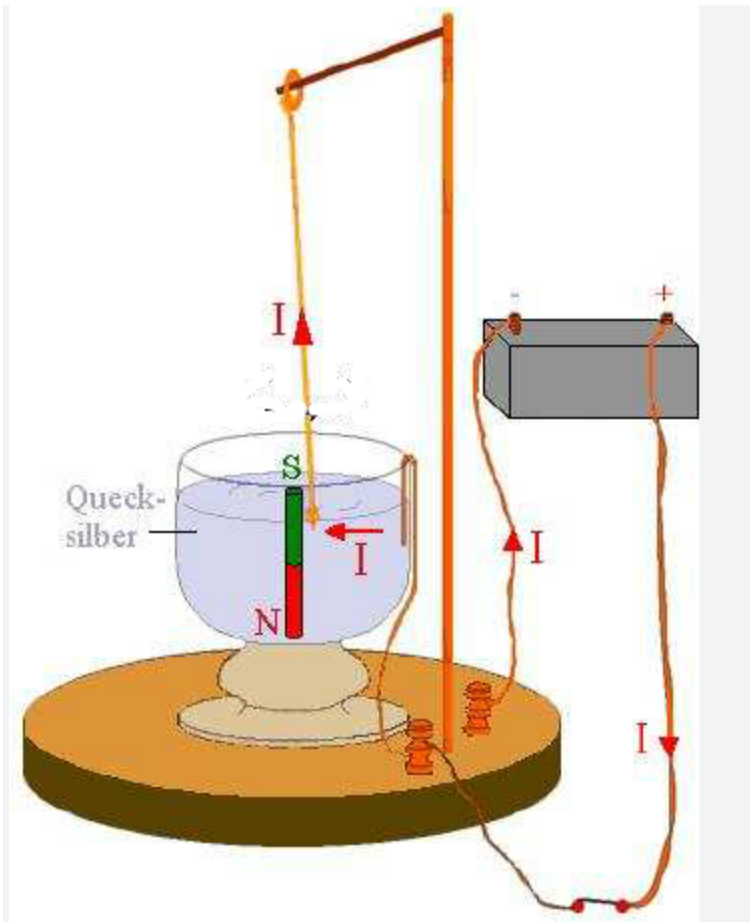
Question 19 5 pts



- ☐ the charge will describe a uniform circular motion. It has a centripetal acceleration. The magnitude of the velocity stays the same (V_x)
- ☐ will accelerate up ($a_y > 0$) while moving at a constant speed (V_x). It describes a parabola curved up
- ☐ the charge has no acceleration and its motion is a line along the positive x-axis. The velocity is constant.
- ☐ will accelerate down ($a_y < 0$) while moving at a constant speed (V_x). It describes a parabola curved down.



Question 20 5 pts



Here is a diagram of the first motor built by Faraday. The yellow solid wire is mobile and will start spinning around the magnet. The liquid is mercury so its a conductor. In the situation illustrated above, the magnetic force acting on the yellow wire inside the liquid is:

hint: I the current is the red arrow in mercury. The magnetic field is produced by a bar magnet. So you know the configuration of the field lines. From ... to

- ☐ the force is @ right
- ☐ the force is in the screen (away from you)
- ☐ the force is out of the screen. Toward you.
- ☐ the force is @ left.

Not saved

Submit Quiz